A Lifecycle-Aware Evaluation Framework for Battery State-of-Health Estimation

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Abstract. Condition-based maintenance of battery energy-storage assets requires State-of-Health (SoH) estimation deployable on the asset itself. This paper establishes a rigorously evaluated gradient-boosted-tree baseline against which edge-native neural architectures can be benchmarked. A reproducible LightGBM regression pipeline is developed on the NASA battery dataset (34 cells, nine protocols, 4–44 °C, 1–4 A constant-current and pulsed discharge); automated quality flagging retains 20 cells for supervised learning. A systematic ablation crosses three feature sets (31, 61, 134 features) with three model configurations under leave-one-battery-out cross-validation at two tiers: same-protocol and cross-protocol. Performance is reported through regression metrics and a four-band maintenance classification (Good, Fair, Poor, Critical). Three findings emerge. First, feature-count ablation produces an inverted-U curve with optimum at 61 features (MAE 0.031, R^2 0.862), degrading beyond this regardless of model capacity. Second, Critical-band recall is weak (7.1 % same-protocol; 5.2 % cross-protocol), concentrating error in the knee-point region most relevant to maintenance. Third, cross-protocol evaluation reveals catastrophic generalisation failure on out-of-envelope cells, making protocol conditioning essential for any deployable successor. Tree-ensemble inference is further shown to be structurally ill-suited to Cortex-M-class targets, while a preliminary 10k-parameter depthwise-separable 1D CNN evaluated under the same harness matches the baseline on same-protocol regression (MAE 0.028) and remains quantisable to microcontroller scale.

Keywords: Battery State-of-Health · TinyML Edge Deployment · LightGBM Baseline · Cross-Protocol Generalisation · Lifecycle Classification · NASA Battery Dataset.

1 Introduction

Battery energy-storage systems are now critical maintained assets across rail, grid, telecommunications and electric-vehicle fleets. The economic case for condition-based maintenance (CBM) of these assets rests on accurate, timely knowledge of State-of-Health (SoH), conventionally the ratio of present usable capacity to initial capacity, because SoH determines both the residual service value of the asset

and the urgency of replacement. SoH also gates operational safety, since ageing raises internal resistance and heat generation, so an over-estimated SoH leaves a cell in service while overheating risk grows, and it is the input to remaining-useful-life (RUL) prognosis, on which replacement decisions are ultimately taken. In distributed deployments (trackside cabinets, remote telecommunication sites, rolling stock), cloud-dependent SoH estimation is unattractive: connectivity is intermittent, latency unbounded, and streaming raw cycle data conflicts with the installations' own energy budgets. SoH estimation must therefore run *on or near the asset*, on microcontroller-class hardware with memory measured in hundreds of kilobytes.

A large literature reports strong SoH estimation accuracy with machine-learned regressors, but three habits limit its usefulness for maintenance engineering. First, evaluation is frequently performed with random train/test splits that allow cycles from the same cell to appear on both sides of the split, leaking cell identity and inflating accuracy. Second, models are almost always trained and tested within a single operating protocol, leaving unanswered the question a maintainer actually asks: *will this estimator work on my cells, under my duty cycle?* Third, regression metrics (MAE, RMSE, R^2) are reported in isolation, although maintenance action is triggered by threshold crossings, not average error; an estimator can post excellent MAE while missing the degradation knee that separates a healthy asset from one needing urgent replacement.

This paper addresses these gaps by building a deliberately conventional model, a LightGBM [9] gradient-boosted-tree regressor, inside an unconventional, lifecycle-aware evaluation framework, and reporting honestly where it fails. The framework is designed so that the same harness can later score edge-native neural successors like-for-like. The contributions are:

1. **A reproducible two-tier evaluation protocol** on the NASA battery dataset [15]: leave-one-battery-out (LOBO) cross-validation at a *same-protocol* and a *cross-protocol* tier, separating interpolation ability from the generalisation actually required in service.
2. **A systematic feature-count ablation** (31, 61 and 134 features) crossed with three model capacities, revealing an inverted-U accuracy curve whose optimum (61 features; MAE 0.031, R^2 0.862) is independent of model capacity: evidence that feature design, not model size, is the binding constraint.
3. **A four-band maintenance classification scheme** (Good $\geq 90\,\%$, Fair 80–90 %, Poor 70–80 %, Critical $< 70\,\%$ SoH) that maps regression output onto operational decisions, exposing a Critical-band recall of only 7.1 % (same-protocol) and 5.2 % (cross-protocol), a weakness invisible to aggregate regression metrics.
4. **A deployability analysis and successor preview**: tree-ensemble inference is structurally ill-matched to Cortex-M-class targets, and a 10,169-parameter depthwise-separable 1D CNN on raw resampled cycles, evaluated under the identical harness, matches the baseline while remaining amenable to INT8 quantisation.

The intent is explicitly that of a *baseline paper*: the framework and the documented failure modes (knee-region insensitivity, cross-protocol collapse) matter more than the headline numbers, because they define what any successor must overcome.

2 Related Work

Data-driven SoH estimation. Machine-learning approaches to lithium-ion SoH and remaining-useful-life estimation span Gaussian-process regression [13], support-vector machines, gradient-boosted trees and deep sequence models [19]; see [3] for a critical review. Feature-based methods dominate practical work [14], extracting health indicators from partial charge/discharge curves, incremental-capacity peaks [18], temperature trajectories and internal-resistance proxies. Severson et al. [17] demonstrated that features from early discharge curves predict cycle life across a large LFP cell population, while knee-point studies [6] establish that degradation is distinctly non-linear, with a late-life knee that dominates replacement timing. The NASA Prognostics Center of Excellence dataset [15] remains the most widely used public benchmark, yet most published results on it use within-cell or random splits, and cross-protocol evaluation is rare. This work treats the knee region and protocol heterogeneity as the primary objects of evaluation.

Evaluation leakage in battery prognostics. When cycles from one cell appear in both training and test sets, the learner can exploit cell identity rather than degradation physics; recent work has documented the resulting information leakage and metric inflation explicitly [7], and best-practice pipelines evaluate cell-wise for the same reason [14]. LOBO cross-validation eliminates this channel by construction. The two-tier extension used here additionally separates two questions that single-tier LOBO conflates: whether the model generalises to an unseen cell *of the same protocol*, and whether it generalises to an unseen *protocol*. Table 1 positions this design against the closest prior practice, from foundational cell-level splits with operating policies shared across folds [17, 14, 7] to the most recent leakage-aware evaluations: Madani et al. [12] benchmark 32 NASA cells (4–43 °C) under cell-wise GroupKFold, and Bayat [2] applies LOBO to the same four NASA cells as Tier 1 (best RMSE 0.034, R^2 0.865; SVR collapses to negative R^2 out-of-cell), yet in all of these the training folds still span every operating condition, so generalisation to an unseen *protocol* remains untested. Here that protocol, not only the cell, is the held-out unit at Tier 2, and the cost of the step is measured on otherwise identical models: MAE rises from 0.031 to 0.049 and Critical-band recall falls from 7.1 % to 5.2 % (Sects. 5.2 and 5.4), a penalty invisible to any single-tier cell-wise protocol.

TinyML and edge inference. The TinyML literature establishes that sub-100 kB neural networks achieve useful accuracy on microcontrollers via frameworks such as TensorFlow Lite Micro [5], CMSIS-NN [10] and system–architecture

Table 1. Validation protocols of the closest prior studies vs. this work.

Study	Held-out unit	Test conditions	Protocol transfer
Severson et al. [17]	cell	policies shared with training	no
Roman et al. [14]	cell (per dataset)	within-dataset	no
Geslin et al. [7]	cell (leakage analysis)	within-protocol	no
Madani et al. [12]	cell (GroupKFold)	mixed 4–43 °C folds	no
Bayat [2]	cell (LOBO)	same NASA cells, one regime	no
This work (Tier 1 / Tier 2)	cell / cell + protocol	9 protocols, 4–44 °C, 1–4 A	yes (Tier 2)

co-design [11], with depthwise-separable convolutions [8] as the canonical parameter-efficiency primitive and temporal convolutional networks [1] as a strong sequence-modelling candidate. Tree ensembles, in contrast, have no comparably mature bare-metal deployment path: inference requires traversal of hundreds of trees with irregular memory access, mainstream runtimes assume an operating system and heap allocation, and published tree-ensemble edge work targets richer IoT-class nodes [4]. This asymmetry motivates Sect. 6.

3 Dataset and Quality Tiering

3.1 Source Data

The NASA battery dataset [15] contains run-to-failure records for 34 commercial 18650 lithium-ion cells (2 Ah rated) aged at NASA Ames under nine operating protocols spanning 4–44 °C, 1–4 A constant-current, multi-level and pulsed discharge, and cut-off voltages of 2.0–2.7 V. Each discharge cycle provides time series of terminal voltage, current and cell temperature, plus the measured discharged capacity. SoH for cell b at cycle n is defined as

$$\mathrm{SoH}_{b,n} = \frac{C_{b,n}}{C_{b,1}}, \tag{1}$$

where $C_{b,n}$ is the capacity measured at cycle n and $C_{b,1}$ the first recorded capacity of the cell.

3.2 Automated Quality Flagging

Raw records contain measurement artefacts that would corrupt supervised labels, so three automated filters are applied per cell: a **capacity floor** (cycles below 0.1 Ah discarded as instrumentation artefacts); a **spurious regeneration ceiling** (cycles with SoH > 1.15 discarded, since apparent capacity *increases* of this magnitude reflect test-bench anomalies rather than electrochemistry); and a **minimum-history floor** (cells retaining fewer than 20 valid cycles excluded entirely, since LOBO evaluation on shorter histories is statistically uninformative). Figure 1 summarises the outcome: 20 of 34 cells are retained and 14 are excluded

Table 2. Excluded cells (14 of 34). Valid = cycles surviving the capacity floor and regeneration ceiling; all excluded cells fall below the 20-cycle minimum.

Cells	Ambient	Raw	Valid	Dominant artefact
B0033, B0034, B0036	24 °C	197	0–1	regeneration ceiling (SoH up to 27.5×)
B0038–B0040	24/44 °C	47	1–10	regeneration ceiling (up to 14.9×)
B0041	4 °C	67	0	first-cycle capacity 0.06 Ah corrupts SoH base
B0049–B0052	4 °C	4–25	1–18	regeneration ceiling + short usable history
B0054–B0056	4 °C	102–103	1–5	regeneration ceiling

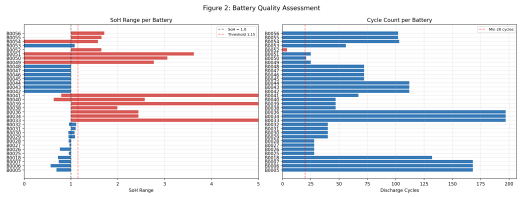


Fig. 1. Automated quality assessment across all 34 cells. Left: per-cell SoH range against the 1.15 regeneration ceiling (cells in red excluded). Right: valid discharge-cycle counts against the 20-cycle minimum.

(the corpus contains no cells B0035 or B0037). Table 2 enumerates the excluded cells and the filter each fails; all 14 exclusions share one mechanism: spurious capacity readings (regeneration artefacts with apparent SoH up to 27.5× initial capacity, or near-zero first-cycle capacities that corrupt Eq. 1) leave fewer than 20 valid cycles.

The criteria therefore key on instrumentation artefacts, not degradation behaviour: no cell is removed for degrading unusually fast or slowly, and every ambient temperature among the excluded cells (4, 24, 44 °C) remains represented among the retained ones. Selection bias against genuine degradation trajectories is thus limited, though the retained population is necessarily biased toward cells whose instrumentation stayed reliable, a caveat shared by any curated benchmark.

3.3 Two-Tier Evaluation Design

The retained cells are organised into two evaluation tiers:

Tier 1, same-protocol. Cells B0005, B0006, B0007 and B0018, all aged under the canonical room-temperature 2 A constant-current protocol. LOBO over these four cells (636 test cycles) measures generalisation to an unseen cell *within* a known operating envelope, the most favourable realistic deployment case.

Tier 2, cross-protocol. All 20 retained cells, whose protocols span 4–44 °C, 1–4 A and differing cut-off voltages. LOBO over the full set (20 folds) measures

generalisation to unseen cells *and* unseen operating regimes simultaneously, the condition an estimator meets when deployed on a fleet it was not trained on.

The design separates interpolation from extrapolation: a model may legitimately be deployed under Tier 1 conditions (fleet-specific training data), and Tier 2 quantifies the penalty when that assumption fails.

4 Methodology

4.1 Feature Engineering and Ablation Design

Three nested feature sets of 31, 61 and 134 features are extracted from each discharge cycle's voltage, current and temperature series, constructed *domain-informed* rather than by automated selection: a 31-feature core of cycle-level summary statistics with energy integrals; a 61-feature set adding degradation-sensitive curve-shape descriptors of known electrochemical relevance; and a 134-feature set adding exhaustive fine-grained segment statistics and channel interactions. Appendix A defines all three sets; the extraction notebook and per-cycle base-feature matrix are available from the authors on request. Crossing the three feature sets with three LightGBM capacity configurations (shallow, deeper, much-deeper) yields a 3×3 ablation grid evaluated identically under both tiers.

4.2 Models

The baseline regressor is LightGBM [9], chosen for state-of-the-art tabular accuracy, native missing-value handling and fast retraining, with early stopping in every LOBO fold. The final configuration (61 features, much-deeper capacity) is the reference model for all results below.

4.3 Lifecycle Classification Scheme

Regression output is mapped to four maintenance bands aligned with operational decision thresholds:

Band	SoH range	Condition	Maintenance action
Good	$\geq 90\,\%$	nominal	routine monitoring
Fair	80–90 %	early wear	increased monitoring cadence
Poor	70–80 %	advanced wear	schedule replacement
Critical	$< 70\,\%$	end-of-life exceeded	urgent replacement

The 80 % boundary corresponds to the conventional end-of-life criterion for lithium-ion cells [16]; the 90 % boundary marks the onset of measurable wear at which operators raise monitoring cadence, and the 70 % boundary the region

where knee-driven accelerated fade [6] and rising internal resistance make continued operation a safety as well as an economic risk. All three boundaries are configurable parameters of the harness rather than claims of universality. Band classification is evaluated by per-band recall and overall band accuracy, computed from the same LOBO predictions as the regression metrics; no separate classifier is trained, because the operational question is precisely whether the *regressor's* output can be trusted at the thresholds.

4.4 Evaluation Protocol

For each tier, LOBO folds are constructed as in Sect. 3.3. Reported regression metrics are the across-fold mean ± standard deviation of per-cell MAE and R^2 (SoH in absolute fraction units, so MAE 0.031 ≡ 3.1 percentage points of SoH). Band metrics are computed by pooling all LOBO test predictions within a tier. No cycle of any test cell contributes to the training loss of its fold, and all fitted preprocessing is fold-local: imputation statistics and, for the successor of Sect. 6, per-channel normalisation constants are computed on each fold's training cells only. One disclosed exception: the early-stopping round is monitored on the held-out cell; this mild optimism applies identically to all nine configurations and both model families, so the reported orderings and negative findings are unaffected.

5 Results

5.1 Feature-Count Ablation: an Inverted U

Under Tier 1, the much-deeper configuration improves from 31 to 61 features (MAE $0.0333 \rightarrow 0.031$; R^2 $0.838 \rightarrow 0.862$) and then *degrades* at 134 features (MAE 0.0335; R^2 0.838), an inverted-U curve. The same ordering holds across all three capacity configurations, indicating that the degradation is not a capacity artefact: beyond the 61-feature optimum, additional features inject noise faster than signal under LOBO evaluation, and no amount of model capacity recovers it. The practical implication for edge deployment is encouraging: the optimal representation is also a compact one.

5.2 Tier 1: Same-Protocol Regression

Under Tier 1 LOBO the reference configuration achieves **MAE 0.031 ± 0.019** and R^2 **0.862 ± 0.102**. Figure 2 shows per-cell predicted-vs-actual trajectories: three of four cells track closely, while B0006, the deepest-degrading cell (terminal SoH 0.57), is systematically over-estimated in late life, the prediction plateauing near 0.74 as the true trajectory continues downward. This is the knee-region failure in its clearest form: the model has never observed degradation that deep and saturates at the edge of its training envelope.

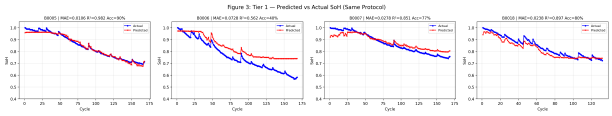


Fig. 2. Tier 1 LOBO predicted vs. actual SoH per held-out cell. Note the late-life plateau on B0006, where predictions saturate near the training envelope boundary while true SoH continues to decline.

Table 3. Per-band recall of the LightGBM baseline under both evaluation tiers (pooled LOBO predictions). n = true cycles per band.

	Tier 1 (same-protocol)		Tier 2 (cross-protocol)	
Band	n	Recall	n	Recall
Good	200	91.5 %	628	83.8 %
Fair	159	81.1 %	283	80.6 %
Poor	207	66.7 %	361	65.1 %
Critical	70	**7.1 %**	174	**5.2 %**
Overall band accuracy		72.0 %		69.0 %

5.3 Lifecycle Band Classification: the Critical-Band Blind Spot

Aggregate band accuracy under Tier 1 is 72.0 %. Figure 3(left) and Table 3 decompose this by band, the central negative result: recall degrades monotonically with severity, Good 91.5 %, Fair 81.1 %, Poor 66.7 %, **Critical only 7.1 %** (5 of 70 identified; 65 mis-classified one band optimistic, as Poor). Errors are almost exclusively *adjacent-band, optimistic*: the regressor under-estimates severity precisely where it matters most, so an asset-management system would schedule replacement for most truly-critical cells (predicted Poor) but lose the urgency signal.

Two mechanisms explain the blind spot: *class exposure* (Critical cycles are the rarest in training, so the objective is dominated by the healthy majority) and *knee non-linearity* (a regressor smoothing over the capacity knee lags the trajectory exactly when it steepens; Fig. 2, B0006). Both suggest concrete remedies (severity-weighted or ordinal objectives, knee-aware features, protocol conditioning) taken up in Sect. 7.

5.4 Tier 2: Cross-Protocol Generalisation Failure

Under Tier 2 LOBO the same configuration posts **MAE 0.049 ± 0.043**: mean error rises by ∼60 %, and the standard deviation reveals the real story: per-cell performance becomes *bimodal*. Cells whose protocols resemble the training mixture degrade modestly; out-of-envelope cells fail catastrophically, with per-cell R^2 frequently negative. The band-level view (Fig. 3, right; Table 3) shows overall accuracy falling to 69.0 % and Critical-band recall to 5.2 %.

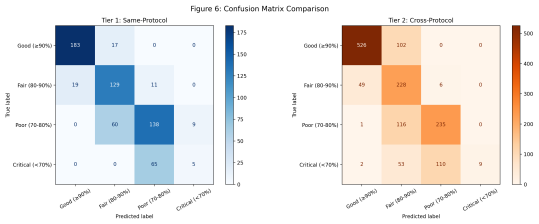


Fig. 3. Lifecycle-band confusion matrices, Tier 1 (left) vs. Tier 2 (right). Errors concentrate immediately below the diagonal (optimistic, adjacent-band), and the Critical row is almost entirely mis-assigned to Poor in both tiers.

The qualitative conclusion outweighs the numbers: **a feature-based tree ensemble trained without protocol information does not transfer across operating regimes**. Features carrying degradation signal under one temperature/load regime alias differently under another; the model cannot distinguish "different protocol" from "different health". A deployable successor must either condition on protocol descriptors (temperature, load, cut-off) or learn representations invariant to them; Tier 2, not Tier 1, is the scoreboard that matters for deployment claims.

5.5 Hardware Feasibility of the Baseline

The reference model artefact serialises to a few kilobytes, which superficially suggests edge readiness; the obstacle is the *inference path*, not parameter storage. Gradient-boosted ensembles require (i) hundreds of sequential tree traversals per prediction with irregular memory access, (ii) a runtime assuming heap allocation and an OS, and (iii) upstream computation of the full 61-feature vector, whose scratch memory rivals the model itself. None of this maps onto a Cortex-M-class target (≤ 256 kB SRAM, no OS, fixed-point preferred), and no mature deployment toolchain exists for tree ensembles comparable to TensorFlow Lite Micro [5] or CMSIS-NN [10]; energy-aware tree-ensemble inference remains open even on richer IoT nodes [4]. The baseline is *structurally* non-deployable on the target class: the benchmark to beat, not the product to ship.

6 Preliminary Successor Evaluation: a 10k-Parameter 1D CNN

This section reports a preliminary like-for-like evaluation of the leading successor candidate: a compact 1D CNN with depthwise-separable convolutions [8], the architecture family with the most mature microcontroller deployment path [5, 10]. Temporal convolutional networks [1] are the planned second candidate.

Table 4. LightGBM baseline vs. preliminary 1D CNN under identical two-tier LOBO evaluation. Band acc. = overall four-band classification accuracy.

Metric	LightGBM (61 feats)	1D CNN (raw cycles)
Tier 1 MAE	0.031 ± 0.019	$\mathbf{0.028 \pm 0.010}$
Tier 1 R^2	$\mathbf{0.862 \pm 0.102}$	0.826 ± 0.160
Tier 1 band acc.	72.0 %	**78.7 %**
Tier 2 MAE	0.049 ± 0.043	$\mathbf{0.040 \pm 0.033}$
Tier 2 R^2	wide, often < 0	-2.13 ± 5.22
Tier 2 band acc.	69.0 %	**77.0 %**
Parameters / artefact	n/a (ensemble)	10,169 / ∼40 kB fp32
Cortex-M path	none mature	TFLM / CMSIS-NN, INT8

Architecture and training. Each cycle's voltage, current and temperature series are linearly resampled to 200 points and per-channel z-scored, a 3×200 input with *no* hand-crafted features. The network comprises a standard convolution (kernel 7, 24 channels), two depthwise-separable blocks (24→48, kernel 5; 48→96, kernel 3) with batch normalisation and max-pooling, global average pooling, and a 96→32→1 head: **10,169 parameters** (∼40 kB float32; ∼10 kB INT8). Training uses AdamW (lr 2×10^{-3}, weight decay 10^{-4}), L1 loss, cosine annealing, batch size 64, early stopping (patience 12, max 80 epochs), seed 42; the full two-tier LOBO evaluation completes in 314 s on a laptop CPU.

Results under the identical harness. Table 4 compares the CNN against the LightGBM reference under the same folds, filters and metrics.

Three observations. First, *the successor class is viable*: on raw resampled cycles with no feature engineering, the CNN matches or slightly improves the baseline on every Tier 1 and Tier 2 aggregate, while being deployable by construction. Second, *the baseline's failure modes persist*: Tier 2 R^2 remains strongly negative on out-of-envelope cells (worst cell: MAE 0.164, R^2 −19.0), confirming that cross-protocol collapse is a property of the problem framing (no protocol conditioning), not of the model family. Third, the deepest-degradation folds remain hardest for both families, reinforcing the knee-region diagnosis. Full treatment of the successor (INT8 quantisation, on-target latency and energy measurement, protocol conditioning, severity-aware objectives) is the subject of the follow-on paper.

7 Limitations and Roadmap

Limitations. (i) The NASA dataset comprises a single cell format from one manufacturer; feature-optimality conclusions are dataset-conditional. (ii) The four-band thresholds, though aligned with the 80 % end-of-life convention [16], vary by application and chemistry; the framework, not the thresholds, is the contribution. (iii) Feature provenance is documented in Appendix A, with the

extraction notebook and feature matrix available from the authors on request. (iv) Tree-ensemble hardware infeasibility is argued structurally, not by an attempted port.

Roadmap. The successor programme follows directly from the three findings: (1) *protocol conditioning*: inject temperature, load and cut-off descriptors (or learn protocol-invariant representations) to attack the Tier 2 collapse; (2) *severity-aware objectives*: ordinal or cost-weighted losses that price a missed Critical band higher than a missed Good band, attacking the 7.1 % recall; (3) *edge-native architectures*: the 1D CNN of Sect. 6 quantised to INT8 under TFLM/CMSIS-NN with on-target latency and energy budgets, and a TCN [1] comparison; (4) *cross-dataset validation* on additional public corpora to decondition the feature-count finding; (5) *safety- and RUL-aware extension*: couple SoH estimation with RUL prognosis (cycles-to-threshold) and thermal indicators (temperature-rise anomalies, charge-cycle stress), so replacement decisions and overheating risk are addressed jointly rather than through SoH bands alone.

8 Conclusion

This paper contributes a lifecycle-aware evaluation framework for battery SoH estimation and a rigorously evaluated LightGBM baseline within it. The framework's value lies in what it exposes: a feature-count optimum (61) that model capacity cannot move; Critical-band recall of 7.1 % (same-protocol) and 5.2 % (cross-protocol) hidden beneath respectable aggregates; and a catastrophic cross-protocol failure that disqualifies naive deployment. A preliminary 10k-parameter 1D CNN evaluated under the identical harness matches the baseline while being structurally deployable on Cortex-M-class hardware, validating the successor path. The baseline, harness and quality-tiering pipeline are reproducible from the public NASA dataset and offered as a common scoreboard, on which Tier 2 performance, not Tier 1 interpolation, is the figure of merit for deployable battery health estimation.

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Appendix A: Feature-Set Definitions

All features are computed per discharge cycle from the terminal-voltage (V), current (I) and temperature (T) series and their timestamps; the three sets are nested. **Base (31):** per-channel mean, std, min, max, range, median, skewness and kurtosis of V, I, T (24), plus discharge duration; dwell time below 3.5 V and 3.0 V; mid-discharge voltage slope $(V_{3n/4} - V_{n/4})/(t_{3n/4} - t_{n/4})$; discharge energy $\sum V|I|$; mean power; and end-to-start temperature rise. These 31 names match the serialised model artefact exactly. **Extended (61 = 31 + 30):** adds per-channel percentiles (p10, p25, p75, p90); discharge-third segment means and per-third voltage drops; dV/dt and d^2V/dt^2 statistics; current coefficient of variation; end-to-end temperature gradient; voltage and

power time-integrals. **Maximum (134 = 61 + 73):** adds five-segment statistics of all three channels; V and T at 5/15/25/50/75/85/95 % of discharge time; dQ/dV statistics (mean, std, max, min, range, skewness, kurtosis); internal-resistance proxies ($V_0 - V_5$, $V_5 - V_{10}$); power std/skewness/kurtosis; half-cycle energy integrals and their ratio; temperature rate-of-change mean/max/std; $|I|$ time-integral; FFT of mean-centred V (peak, peak frequency, energy, first five components) and I (peak, energy); V–I, V–T, I–T correlations; and $V \cdot T$ product mean and std.

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